

# Trade Analysis Report

## 1. Introduction

This report outlines the methodology, findings, and assumptions used in analyzing historical trade data from Binance accounts over 90 days. The objective is to calculate key financial metrics, rank accounts based on performance, and provide a list of the top 20 ranked accounts.

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## 2. Methodology

### 2.1 Data Cleaning & Preprocessing

- **Loaded trade dataset** and inspected for missing or malformed values.
- **Ensured correct data types** (e.g., numeric conversions for quantity, qty, and realizedProfit).
- **Classified trades** by combining side and positionSide to distinguish between long\_open, long\_close, short\_open, and short\_close.

### 2.2 Metrics Calculation

For each account (Port\_IDs), we calculated the following metrics:

#### 1. Return on Investment (ROI %)

- Formula:
- Measures profitability relative to total capital invested.

#### 2. Profit and Loss (PnL)

- Formula:
- Represents the net earnings or losses across all trades.

#### 3. Sharpe Ratio

- Formula:
- Evaluates risk-adjusted performance.

#### 4. Maximum Drawdown (MDD %)

- Formula:
- Captures the worst peak-to-trough decline.

#### 5. Win Rate (%)

- Formula:
- Measures consistency in profitability.

#### 6. Win Positions & Total Positions

- Win Positions: Number of trades with positive realized profit.
- Total Positions: Count of all trades.

## 2.3 Ranking Algorithm

A weighted scoring system ranks accounts based on:

- **ROI (40%)**
- **PnL (30%)**
- **Sharpe Ratio (20%)**
- **Win Rate (10%)**
- **MDD (-10%)** (*negative weight as higher drawdown is unfavorable*)

The performance score is computed as:

The top 20 accounts with the highest scores are extracted.

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## 3. Findings

- **Higher ROI and Sharpe Ratios** generally correlate with more profitable accounts.
- **Lower Maximum Drawdowns (MDD)** indicate better risk management.
- **Win Rates above 50%** often lead to higher rankings, but ROI and PnL still play dominant roles.
- **Some accounts had strong PnL but poor Sharpe Ratios**, suggesting high risk exposure.

The **top 20 accounts** have been exported to `top_20_ranked_accounts.csv` for further analysis.

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## 4. Assumptions

1. **Trade data is accurate** and free from errors.
  2. **Realized profit correctly reflects trading success**, even if positions remain open.
  3. **Market conditions remain stable** over the analysis period.
  4. **Sharpe Ratio assumes daily returns** and uses an annualization factor of 252.
  5. **Investment is approximated by quantity** (capital allocated per trade).
  6. **All trades are independent** and not affected by external hedging strategies.
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## 5. Conclusion & Next Steps

- The ranking system provides a **quantitative framework** to evaluate account performance.
- Further refinements may include **alternative weighting schemes** or **additional risk-adjusted metrics**.

- **Visualization tools** (e.g., PnL trends, trade distributions) can enhance insights